

2018-Ph-H1-Q9

Section: Particles and Waves

Topic: Spectra

Which of the following causes line emission spectra?

- A. Electrons falling to lower energy levels
- B. Electrons moving to higher energy levels
- C. Protons becoming neutrons
- D. Neutrons becoming protons

Solution:

Line **emission** spectra occur when electrons drop to lower energy levels and emit photons.

Answer: A

Revision Tip:

"Emission" = energy given off → electrons fall to lower levels.

"Absorption" = energy taken in → electrons jump to higher levels.